ISC2 SECURITY CONGRESS 2026

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A Quantum Cybersecurity Platform to Protect Organizations That You Can Use Today

ISC2 Security Congress 2026
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AUDIENCE POLL

POLL

How familiar are you with quantum computing?

1. Completely new to it
2. Mostly familiar with the encryption threat
3. Familiar with quantum computing applications
4. Have tried quantum software or hardware

Live Content Slide

When playing as a slideshow, this slide will display live content

Poll: How familiar are you with quantum computing?

About us

Undergraduate at Rensselaer Polytechnic Institute

Quantum algorithm, ML, cybersecurity research at:

- NSA
- RPI
- Chinese University of Hong Kong (CUHK)



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About us

Director, Monmouth University Cybersecurity Research Center

Research quantum cybersecurity

ISC2 Security Congress speaker

CTF coach



Q-Day could ruin your day

A sufficiently powerful quantum computer
could break widely used public-key cryptography.

“Store now, decrypt later”.
We need to begin the post-quantum migration today.

The timing is uncertain. The need to prepare is real.

We told the Q-Day story last year

Preparing for quantum attacks

How do we protect systems against future quantum attacks?

Post-quantum cryptography

Using quantum for offense?

How capable are current quantum computers?

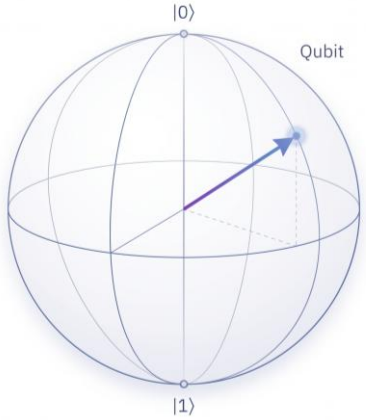
Quantum computing for security analysis

What about defense? Can we use quantum?

We wanted to find out.

So we ran cybersecurity experiments
on a real quantum computer.

Quantum in One Minute



Quantum computers process information using qubits

Qubits can exist in a combination of 1 and 0 (superposition)

So, we're dealing with **probabilities**. E.g., we may be 50% likely to see a qubit as 0 and equally likely to see it as 1.

We apply operations (gates) to control these probabilities.

We can encode data features into qubit states, then use measurement outcomes to compare patterns in the data and perform classification.

Recent Quantum Cybersecurity Research

Goal	Uses
Certified quantum randomness	Verified unpredictability for cryptographic keys https://doi.org/10.1038/s41586-026-10521-8
Private and verifiable quantum cloud computing	Run computations on remote QCs while keeping data private and checking correctness https://doi.org/10.1103/PhysRevLett.132.150604
Quantum-assisted threat detection	Combine with classical ML to assist intrusion & malware detection https://doi.org/10.1016/j.cose.2025.104341

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Last Year: Classifying Network Traffic Data

85%

average hardware accuracy

Benign or malicious NIDS data
Quantum kernel + conventional SVM
8-qubit feature vectors

Four attack types
60 samples per attack, 10 hardware runs

The lesson: this was doable on real hardware.

Cotrupi & Callahan (2025) DOI: 10.1609/aaaiss.v7i1.36903

It worked.
Getting started was hard



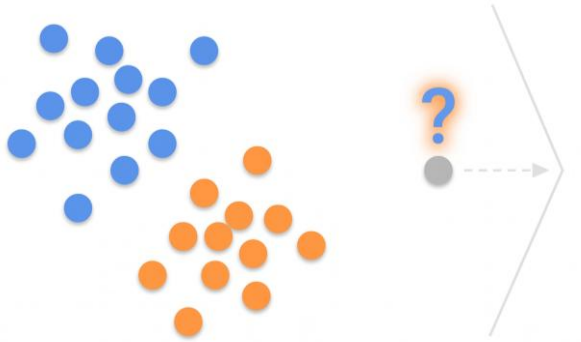
This year, we wanted to tackle a new problem:

Can we detect malware too?

Machine learning for malware classification:

Malware also offers complex data for us to analyze. We can learn and compare patterns in this data to classify as malware or benign.

Quantum is being explored as a potentially better way of comparing those patterns, using a feature map and quantum kernel.



We also wanted to know…

How much does hardware noise affect malware classification?

We compared quantum simulations, real hardware, and conventional machine-learning models.

We measured classification performance and changes in the quantum similarity estimates.

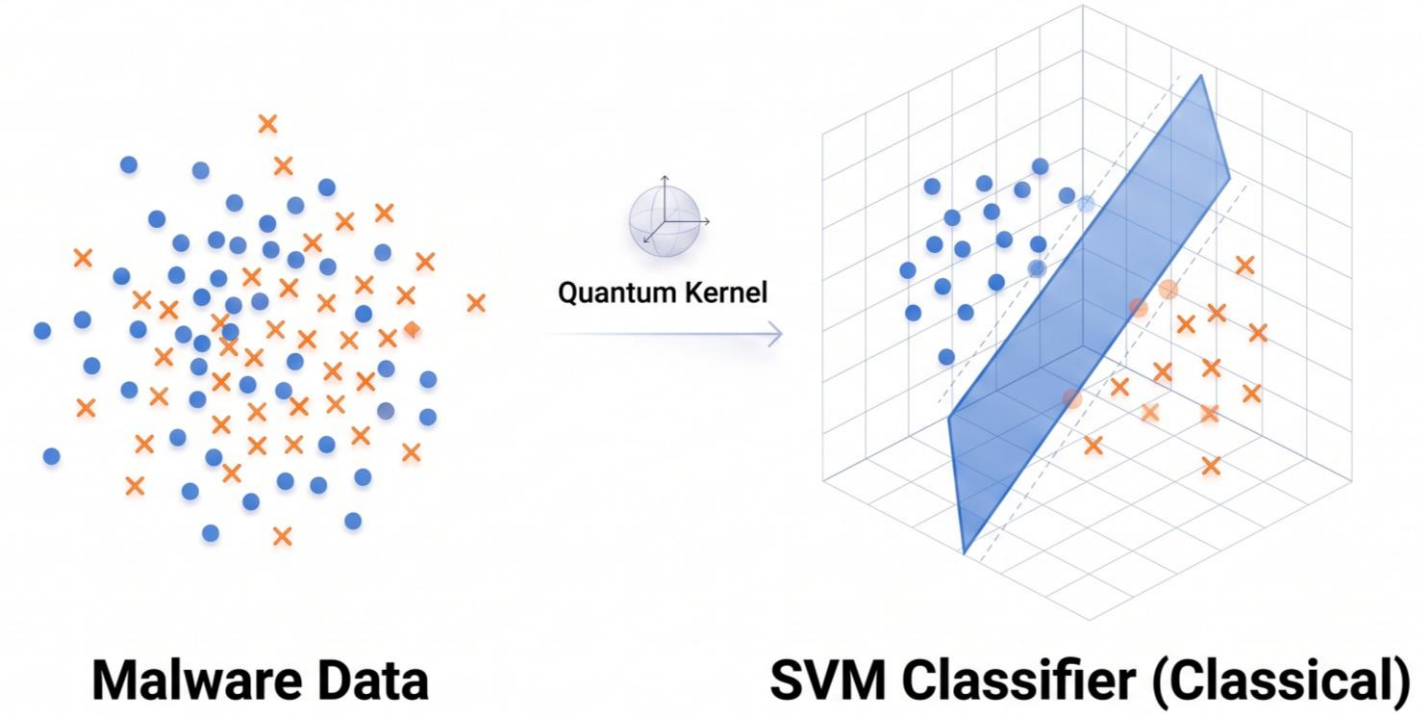
The Quantum Computer We Used

IBM Quantum System One (127 qubits)
Rensselaer Polytechnic Institute



The Methodology

1. Prepare the data features (from malware files)
2. Use quantum circuits to calculate similarities (quantum kernel)
3. Train a conventional classifier (SVM)
4. Predict malware or benign



Preparing the file features

Each sample becomes a list of numbers.

Remove duplicate rows and clean the features.

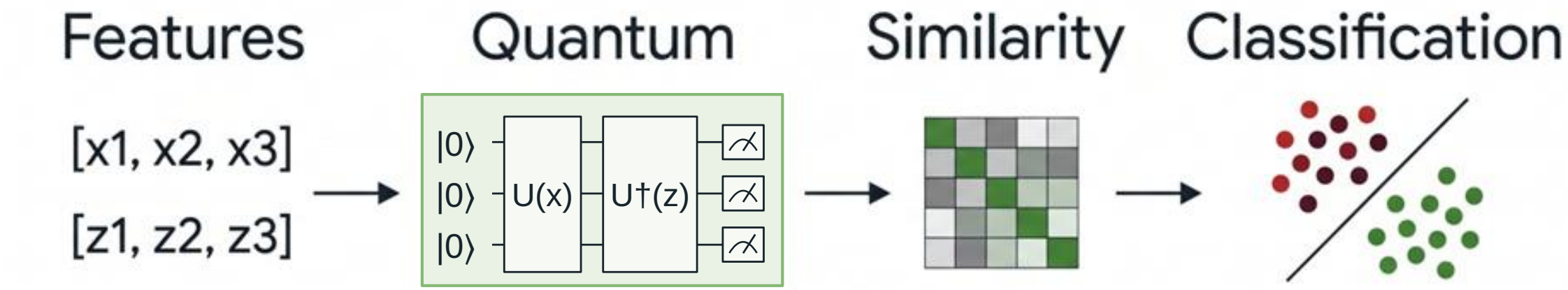
Keep test samples separate.

Fit preprocessing on training data only.

Representation	Qubits
CLaMP native features	10
MBR native features	8
PCA versions	4

PCA compresses the feature list. Here, one feature uses one qubit.

Quantum-calculated similarities with classical predictions



512 measurements per similarity.
Example: 256 / 512 = 0.50.

The SVM uses similarities to predict benign or malware.

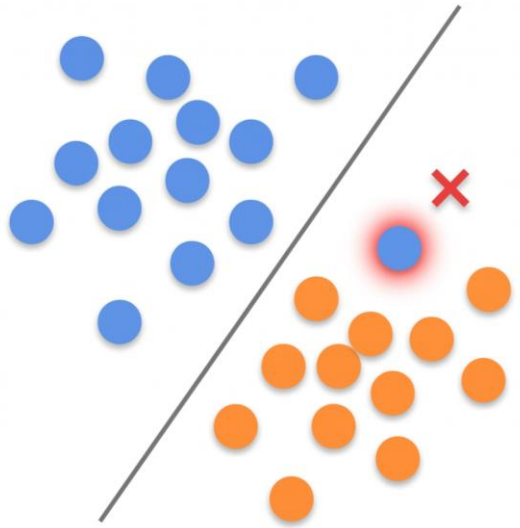
Similarity is not a malware probability.

The classical classifier makes the decision

Stage	Quantum calculation	Conventional computation
Training	Compare the 80 training samples with each other.	The SVM learns from similarities and known labels.
Testing	Compare 80 test samples with the 80 training samples.	The SVM predicts benign or malware.

Each stage uses an 80 × 80 similarity matrix in this study.

What changes on a real quantum computer?



We get errors.

Due to environmental noise and imperfect operations,

qubits can give different results than we expect.

These errors can change which data the classifier gets right.

With a real QC, we can measure these effects and test ways to reduce them.

Four comparisons

Calculation method	
Exact calculation	The mathematical reference
Ideal simulator	What changes when we estimate results from repeated measurements
Noisy simulator	What changes when we add modeled device errors
IBM hardware	What happens on the actual machine

What data did we test on?

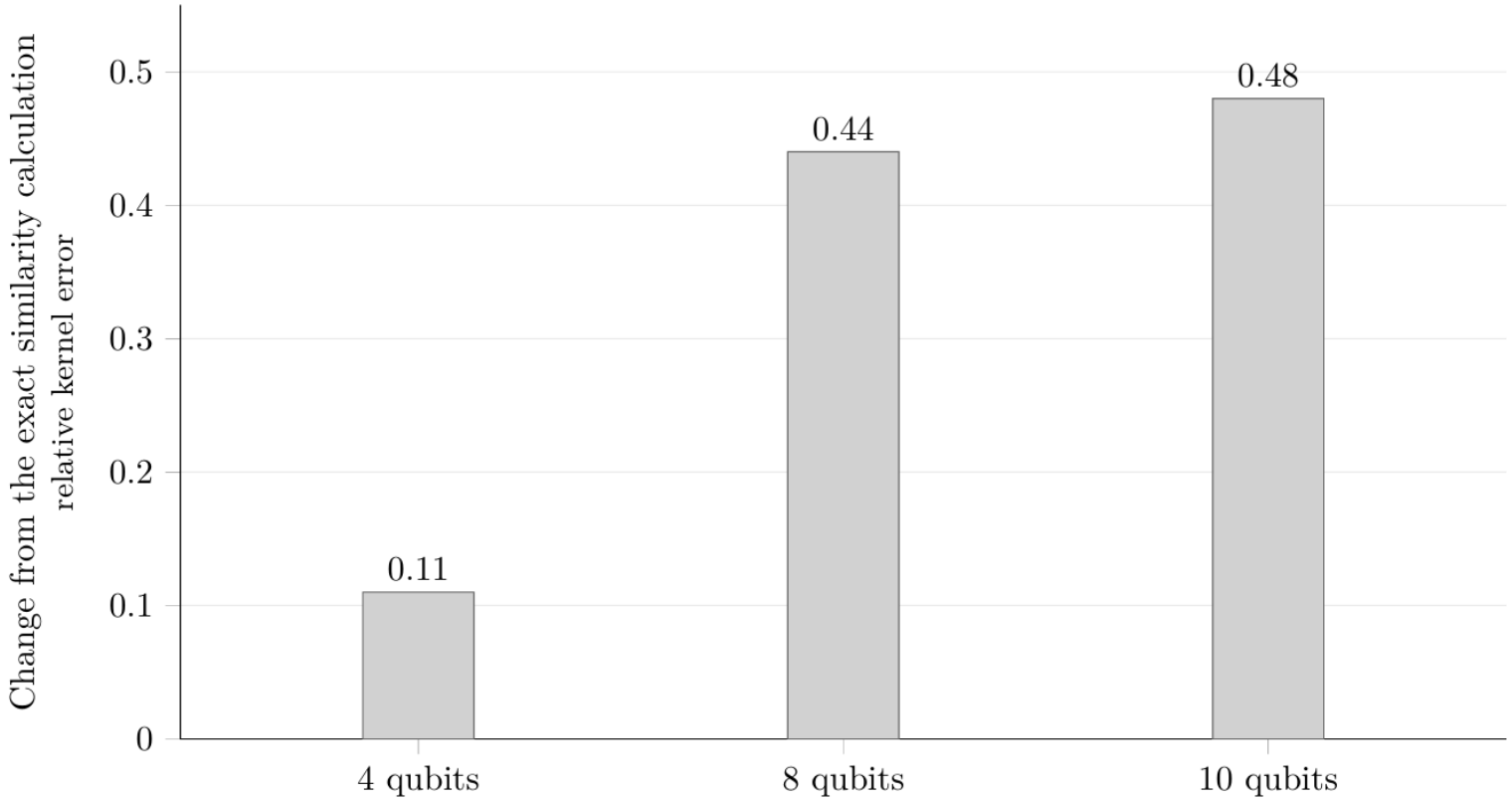
Datasets selected from cybersecurity students at Monmouth University and Rensselaer Polytechnic Institute

Dataset	Platform	Files
CLaMP	Windows	Executables
MBR	Windows	Executables
EMBER2018	Windows	Executables
CIC-YNU-IoTMal	Linux	Sandbox IoT

80 training samples, 80 test samples, 3 repeated splits, equal benign & malware

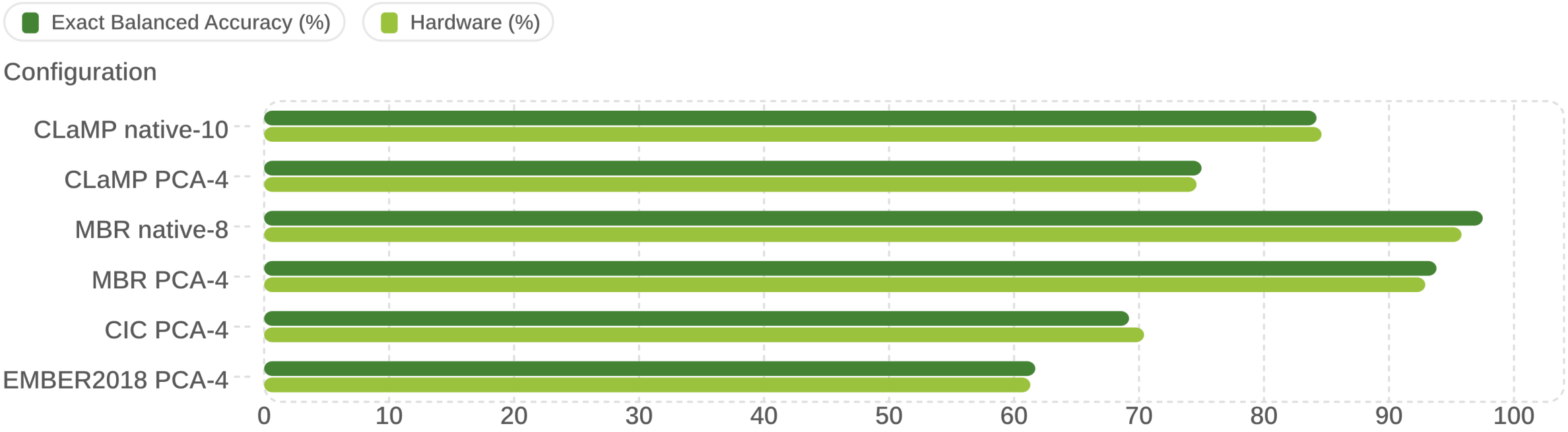
The result: noise changed the similarity measurements

The 8- and 10- qubit circuits showed larger changes than the 4-qubit circuits



But classification accuracy stayed almost the same

Hardware and exact calculated results differed by < 2 percentage points in every config.

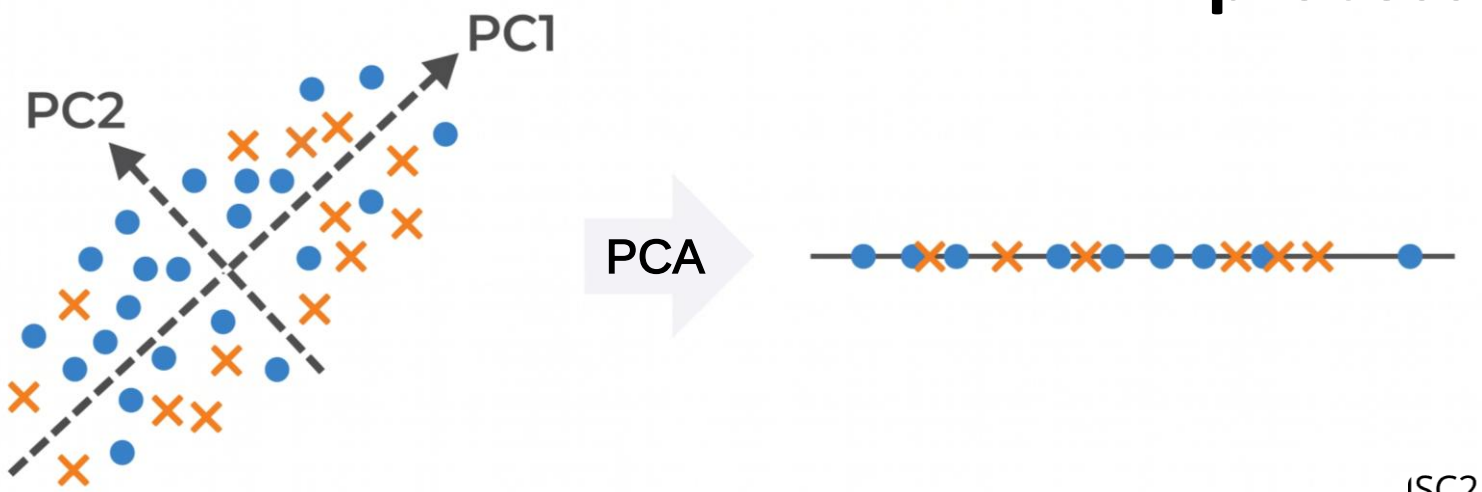


Can we reduce the noise?

Yes, by simplifying the circuit and features.

PCA: Reduce feature space and directly the number of qubits needed.

<u>BUT:</u> We will likely lose some classification accuracy in the process.



Principal Component Analysis

Did it beat conventional machine learning?

No consistent accuracy advantage.

- Classical models matched or exceeded quantum performance on three datasets.
- Quantum models scored higher on MBR.
- We found no consistent accuracy advantage

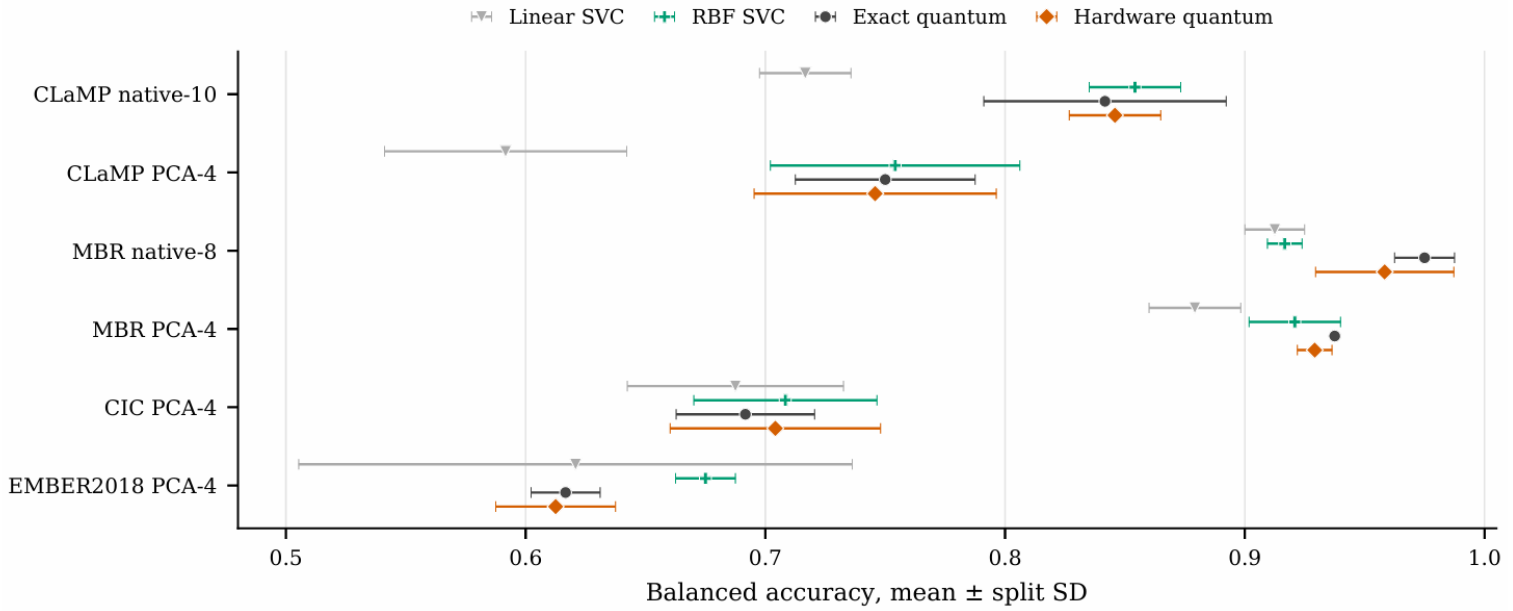


Figure 3: Matched linear-SVC, RBF-SVC, exact-quantum, and hardware-quantum balanced accuracy. Points are three-split means and bars are split standard deviations. Every classifier used $C = 1$ without hyperparameter tuning.

Does this make your job easier today?

Maybe not yet. That’s OK.

Practical advantage at a useful scale remains an open question.

BUT there is still potential for benefit.

Future tests include:

- More demanding tests
  - Larger datasets, more varying threats
- Stronger comparisons
  - Further tune the models to achieve better performance
- Practical performance
  - Evaluate misclassifications in more detail in addition to runtime and cost reports

How would you evaluate a quantum security claim?

What actually ran?

- A simulator, quantum hardware, conventional algorithm?
- Was noise and circuit depth reported?

What was the comparison?

- State-of-the-art classical methods?

How reliable is the result?

- Repeated tests? Reported uncertainty? Reproducible?

We made it easier to get started

Last year

We showed that quantum classification could run on real hardware.

Turning the experiment into your own workflow still took work.

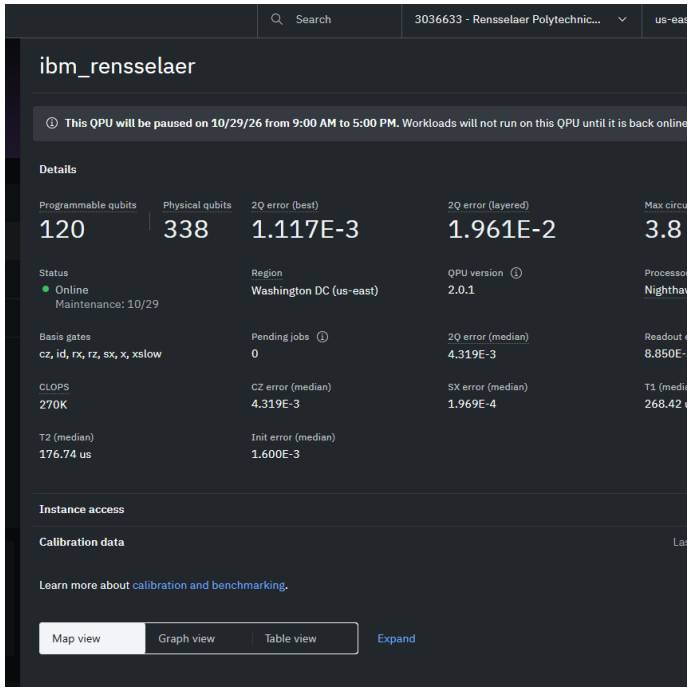
This summer

We shared the malware-classification workflow in a GitHub repository.

You can clone the code and begin with a small Qiskit Aer simulation.

You now have a starting point for your own experiments.

Your first quantum experiment



github.com/ibara/SRP2026

1. Follow the repo setup instructions.
2. Start with a small dataset of file features.
3. Run the quantum-kernel simulator and a conventional SVM.
4. Compare predictions on the same test samples.

This can be run with Qiskit's Aer simulator, or on real IBM hardware via the cloud.

So… quantum is something defenders can use today

You can experiment now.
A laptop simulator is a starting point.

Real hardware can run these small security experiments.
A practical advantage still needs evidence.

Final Takeaways

Mary: Quantum is changing every day. We give a glimpse into the exploratory phase that quantum research is in today and show how it can influence the future of computing and thus cyber defense.

Brian: Quantum for defenders is the exciting part of quantum. It is the part we want you to want. We will continue exploring avenues for quantum to enhance your workflows. Please try our code and tell us what you think ☺

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Questions? Thank You!



Brian Robert Callahan, Ph.D., ISSMP, CISSP, CISM

Specialist Professor, Dept. of Computer Science and Software Engineering; Director,...



Mary Cotrupi

CS @ RPI | Quantum Researcher

References: our work

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